

4 Reductionism

Part 2: Synthetic Reductionism and Empirical Moral Knowledge

4.4 Synthetic reductionism

Synthetic Reductionists appear to escape the Open Question Argument.²⁹ They maintain that there is some true statement of the form, 'For a thing to be good is . . .', where the ellipsis is filled in using non-evaluative terms. But they admit that as Moore showed, what fills in the ellipsis will not be *synonymous* with 'for it to be good'. Consider some analogous cases:

$\angle$ For a thing to be hot is for its molecules to have high average kinetic energy. $\angle$

$\angle$ For a substance to be water is for it to consist of H_2O molecules. $\angle$

$\angle$ For a thing to make a sound is for it to cause a compression wave in the surrounding medium. $\angle$

Quibbles aside, all of those statements are true. But they cannot be verified solely by considering the meanings of the words 'hot', 'water', and 'sound' in English. People knew these words and used them in the same sense we presently do for centuries without knowing the facts expressed by the above statements. Very specialized observations and scientific reasoning were required to confirm those facts. Just so, says the synthetic reductionist, we will someday find (if we have not already found) a theory of what it is for something to be good, and that theory will not be known simply on the basis of the meaning of the word 'good' in English. Instead, it will be known in a way analogous to how the above statements are known to be true, by a kind of inference to the best explanation.³⁰

In this section, I argue that the synthetic reductionists' account of moral knowledge fails: no theory about the nature of goodness can be known in any manner analogous to how the above theories of heat, water, and sound are known. In the end, the synthetic reductionist will have to appeal to ethical intuition, just as the intuitionists do. This deprives their position of one of its central alleged advantages.

4.4.1 Can moral facts be known by observation?

The analogy just drawn between ethical reductionism and various scientific theories plays a fairly serious role in motivating synthetic reductionism. But the analogy fails at a crucial point.

Suppose we are speaking to a synthetic reductionist who holds that for a thing to be good is for it to be N , where N is some natural property, or for short:

$\text{Good} = N$

The reductionist says this is analogous to the theories mentioned above, which I shall abbreviate as follows:

- Heat = Molecular kinetic energy.
- $\text{Water} = \text{H}_2\text{O}$.
- Sound = Compression waves.

In each of these cases, we have two different concepts that refer to the same phenomenon (the concept 'water' is different from the concept H_2O , but there is one thing in reality that both these concepts pick out). In each case, the concept on the left hand side is observational (we can apply it to a thing on the basis of observation), while the concept on the right hand side is theoretical (we apply it to a thing on the basis of inference, including an inference to a scientific theory) and is taken to explain the nature of the phenomenon in question. Thus, we knew that things were hot, initially, because we could feel heat with our hands. We know that things have molecular kinetic energy on the basis of complex inferences from observational data to the molecular theory of matter, and so on. The idea of molecular kinetic energy is thought to explain the underlying nature of heat. Similarly, we know that there is water on the basis of casual observation, but we know that there is H_2O on the basis of complex scientific reasoning. And ' H_2O ' explains the underlying nature of water.

Now, in the reductionist's analogy, which term is 'good' supposed to be analogous to: the term on the left hand side of the '=', or the term on the right? Is 'good' supposed to be like 'heat' or like 'molecular kinetic energy'? Initially, it seems that 'good' is supposed to be analogous to 'heat', because the reductionist's theory is supposed to explain the underlying nature of goodness in terms of natural properties, just as the scientific theory explains the underlying nature of heat in terms of motions of molecules. Certainly ethical reductionists do not take themselves to be explaining the underlying nature of some natural properties in terms of moral properties.

This suggests that moral concepts should be observational, just as 'heat', 'water', and 'sound' are. The reason why we believe, for example, that H_2O is, roughly, that (i) we have independent (that is, pre-theoretical), direct awareness of the presence of water, and (ii) the theory that this substance of which we are aware is composed of H_2O molecules helps explain many of its observable features (again, features of which we know independently of our scientific theory). If we are to take the synthetic reductionist's analogies seriously, then, we should say that we have independent, direct awareness of the presence of goodness, and that the theory that goodness is identical with N helps explain many of the observable features of goodness.

The problem is that in fact, moral properties are entirely unobservable. Moral value does not look like anything, sound like anything, feel (to the touch) like anything, smell like anything, or taste like anything. The only plausible way to maintain that we have direct awareness of moral facts would be to appeal to either 'ethical intuition' or a 'moral sense'—which is precisely the sort of 'mysterious', 'spooky' thing the reductionist wants to avoid.

Gilbert Harman discusses an alleged example of moral observation. You see a group of hoodlums pour gasoline on a cat and ignite it. You immediately think that what they are doing is wrong. According to Harman, 'you do not need to figure anything out; you can see that it is wrong'. But this quotation is misleading by itself; Harman does not think that the wrongness of the action somehow modifies the light rays traveling to your eyes, nor does he think that your alleged ability to see wrongness ultimately explains your moral knowledge. Rather, he thinks that you have background moral beliefs, or a moral sensibility, which cause you, upon seeing an event such as the one described, to immediately (without thinking about it) form the belief, 'That is wrong'. Since the 'seeing' of wrongness depends upon prior moral beliefs, it does not aid in the ultimate explanation of how we know moral truths.

Why does Harman consider this a case of observation at all? He seems to think that to 'observe' that p is just to form a belief that p as a fairly direct result of perception.[33] This is mistaken; a belief formed as a fairly direct result of perception need not be an observation. Suppose I am superstitious and whenever I see a black cat, I immediately (without thinking about it) believe that an airplane is going to crash the next day. I am now seeing a black cat. Do I thereby observe a plane crash? No. More is required for observation. For someone to observe that an object is F , where F is some property, there must be a way that F things look (or sound, smell, etc.), and the object must look (sound, smell, etc.) that way.[34] For instance, I have a pair of artificial leather shoes. Most people can perceive by touch that they are not leather, because there is a way that real leather feels, and these shoes do not feel that way. The point of interest here is that there is no such thing as the way that wrongful actions look or the way that permissible actions look. That is why you cannot literally see, with your eyes, that an action is wrong.[35]

Consider another argument for the possibility of moral observation. Michael Moore argues that since moral properties are natural properties (per his ethical naturalist stance), and since we may be able to perceive the relevant natural properties, we can therefore perceive moral properties. In short, Moore takes the following argument to be valid:

I can perceive N .

$\text{Good} = N$

Therefore, I can perceive goodness.[36]

Before addressing this, I must make a small correction: the relevant question is not whether one can perceive goodness; it is whether one can perceive that a particular thing is good. These are two different questions, and only the latter bears on the problem of moral knowledge. To illustrate, compare this case: I can see your car (your car reflects light to my eyes, and so on) without being in a position to know that the car is yours. If I could see that it is your car, that would enable me to know that it is yours, but merely seeing your car doesn't give me that knowledge. So let us modify Moore's argument as follows:

I can perceive that x is N .

$\text{Good} = N$

Therefore, I can perceive that x is good.

If this inference were valid, Moore would have a viable account of how we can know moral truths. But in fact, the inference is invalid. It has the same form as the following inference:

I can perceive that x is hot.

Heat = Molecular kinetic energy.

Therefore, I can perceive that x 's molecules have high average kinetic energy.

Here, I use one of the reductionists' examples against them. The first two premises may be true, but the conclusion is certainly false. For instance, one can perceive by touch that a stove is hot. But one does not, merely by feeling it, perceive that the stove is composed of fast-moving molecules (even though that is really what it is for the stove to be hot). One has to do some sophisticated theorizing to find that out. Similarly, even if $\text{Good} = N$, and even if one can perceive that x is N , one does not therefore perceive that x is good.³⁷

Value facts, then, are unobservable. This is a problem for the synthetic reductionist's analogies: we were able to discover that heat = molecular kinetic energy, only because we could first identify which things were hot by observation. If we have no pre-theoretical knowledge of which things are good, then we cannot discover that $\text{Good} = N$ in any analogous way. The situation would be like the following scenario: someone tells you that there is an unobservable property called 'torfness'. You are given no initial information about the nature of torfness or which things in the world, if any, are torf. You are told only that torfness is reducible to some scientific property. Your assignment: figure out which scientific property torfness is reducible to.

It seems clear that you do not have enough information to carry out the assignment. Likewise, with no initial information about the nature of goodness or which things have it, we would have no basis for framing any hypotheses as to what natural property goodness might be reducible to. As we have just seen, observation furnishes us with no such initial information. Nor, according to the naturalists, does intuition. One might be tempted to propose that we simply rely on our common sense beliefs about good and bad. But unless we are justified in thinking these beliefs are generally true, this would be arbitrary, and we have as yet seen no account of why those beliefs are justified. Moreover, if the naturalist could give an account of our initial knowledge of the nature of value or of which things are good, he wouldn't need the story about inference to the best explanation that naturalists have been at such pains to develop (see subsections 4.4.2–4.4.4).

4.4.2 Can moral facts be known by inference to the best explanation?

Some reductionists have argued that we are justified in believing moral claims because they provide the best explanations for certain non-moral facts of which we have independent awareness. For instance, the best explanation for why Hitler ordered the Final Solution is that Hitler was depraved; only a depraved person would do such a thing.³⁸ Furthermore, since we know empirically that Hitler ordered the Final Solution, and our justification for believing he was depraved is that that explains why he ordered the Final Solution, our knowledge that he was depraved is also empirical. In short:

1. Moral claims help explain observable facts. For instance, Hitler's depravity helps explain why he ordered the Final Solution.
2. If a claim helps to explain observable facts, then we have empirical justification for believing that claim.
3. Therefore, we have empirical justification for believing some moral claims, such as that Hitler was depraved.

The conclusion in (3) is to be understood as implying that we can have purely empirical justification for believing some moral claims—that is, justification that does not ultimately depend, even partly, upon intuition or any other non-observational source of information.

Most of the discussion in the literature centers on premise (1). There are two main arguments for (1). I call them 'the causal argument' and 'the counterfactual argument'.

The causal argument holds that for x to explain y is for x to be (part of) the cause of y . Hitler's ordering of the Final Solution was caused by certain psychological traits of his, such as extreme racism, hatred, and a lack of respect for

human life. According to the ethical reductionists, these traits constituted Hitler's depravity. Therefore, Hitler's depravity caused his ordering of the Final Solution. (This argument employs the valid principle that if CA causes CB , and BA is identical with CA , then CB causes CB .) Therefore, Hitler's depravity explained his order.³⁹

The counterfactual argument holds that CA explains CB (at least in part) provided the following is true: if CA had not occurred, CB would not have occurred. Now, if Hitler had not been depraved, he would not have ordered the Final Solution. Since moral facts are reducible to natural facts, if Hitler had not been depraved, he would have to have had different psychological traits from those he actually had—for instance, he would have to have not hated the Jews, or have had more respect for human life, etc. And if that were the case, then he would not have ordered the Final Solution. So his depravity explains (at least in part) why he ordered the Final Solution.[40]

I agree that moral facts can be explanatory; premise (1) is true. But premise (2) is false. To see why, return to the reductionists' favorite analogy: the theory that H_2O . Imagine a scientist attempting to justify that theory by the means suggested by ethical reductionists. Suppose Lavoisier had published a paper arguing that we are justified in believing that water is composed of hydrogen and oxygen, on the grounds that the presence of such a compound explains certain observable facts. For instance, I have an observable stain on my coffee table that was caused by water. Since H_2O , we can also say that the stain was caused by H_2O . Therefore, H_2O explains an observable fact. Therefore, we have empirical justification for believing in H_2O .

Here is another argument. If H_2O wasn't spilled on my coffee table, there wouldn't be a water stain there now. For remember, water consists of H_2O , so in order for H_2O not to have been on my coffee table, there would have to have been no water on my coffee table. And if that were the case, there would not be a water stain there now. Thus, the (past) presence of H_2O explains a currently observable fact. So again, we are justified in believing in H_2O .

Thankfully, this is not how scientists reason. Here is part of why scientists believe H_2O : if you apply an electric potential to a sample of water, bubbles of hydrogen form at the cathode. Bubbles of oxygen form at the anode. The mass of water present decreases by the same amount that the mass of hydrogen and oxygen present increases.[41] Furthermore, two parts hydrogen are formed for every one part oxygen (by moles). That is just one of several relevant pieces of experimental evidence, but that will do for illustrative purposes. The moral theorist has no empirical evidence to offer that is analogous to that.

This example suggests that the causal and counterfactual arguments do not show how moral claims can be justified, since analogous arguments would not work for scientific claims. But exactly what is wrong with those arguments?

Begin with the causal argument. If ethical reductionism is true, then moral properties can cause observable effects. Nevertheless, this would not explain how we know moral claims. For us to know a moral claim on the basis of inference to the best explanation, the moral claim would have to explain some observable fact that could not otherwise be explained. For instance, we can know that Hitler was depraved by inference to the best explanation, if the claim that Hitler was depraved provides an explanation for some fact that cannot be explained without invoking moral claims. But in fact, the hypothesis of Hitler's depravity does not explain any new facts about his behavior that aren't already explained by non-moral claims about him, such as that he hated Jews, that he had a lust for power, that he lacked respect for human life, and so on. Since the truth of non-moral claims of that kind is not in dispute, the moral explanation is superfluous and provides no justification for adopting any moral theories. This is not to say, of course, that we have no justification for our moral claims about Hitler; if we were already justified (perhaps on the basis of moral intuition) in accepting moral theories such as that mass murderers are evil, then we can infer that Hitler was evil. The point is that neither our general moral theories nor our moral judgments about Hitler can be established solely on the basis of observation.

The same applies to the counterfactual argument. Given that racial hatred and so on constitute depravity, it is true that had Hitler not been depraved, he would not have acted as he did. But we can explain Hitler's behavior equally well without saying that racial hatred and so on constitute depravity: had Hitler lacked racial hatred, power lust, and so on—whether or not these things constitute depravity—he would not have acted as he did. So moral assumptions add no explanatory power.

To confirm this diagnosis, recall the water/ H_2O example. The water stain on my coffee table fails to confirm the past presence of H_2O , since the hypothesis that H_2O was spilled on the table does not explain any new fact that was not already explicable merely by the hypothesis that water was spilled on the table. But the electrolysis experiment confirms that H_2O is present, because the hypothesis that H_2O is present explains where the hydrogen and oxygen are coming from. That is not explained merely by the hypothesis that water is present, without assuming that water is composed of hydrogen and oxygen.

***4.4.3 Can moral claims be tested?**

Perhaps moral claims are justifiable because they make empirical predictions that can be tested. For instance, the two moral claims 'Hitler was an admirable person' and 'No admirable person would kill millions of people' together entail the following prediction: 'Hitler would not kill millions of people'. This prediction is testable and, as a matter of fact, false. So we see that moral claims are testable just as scientific theories are.

In case you are tempted to object that neither moral claim by itself entails any testable predictions, Nicholas Sturgeon reminds us that it is also true that individual scientific claims do not, in general, entail any testable predictions by themselves; normally a scientific theory makes a testable prediction only given a large background of other scientific knowledge.⁴²

Even granting the last point, it does not seem that this accounts for moral knowledge. Imagine that Lavoisier had argued analogously: Collections of claims about H₂O entail testable predictions. For instance, the pair of claims 'Water = H₂O' and 'H₂O boils at 212°F' together entail the testable prediction, 'Water boils at 212°F'. Since the latter prediction is in fact true, my H₂O beliefs thereby receive confirmation.

The problem is that not all predictions made by a theory create ways the theory can be usefully confirmed. If each prediction is made with the help of a separate ad hoc posit, then the theory's success at prediction will be unimpressive. Thus, the 'Water = H₂O' theory can make a series of testable predictions about the nature of water, if we add to it the ad hoc suppositions that H₂O would be a clear, odorless liquid, that it would boil at 212°F, and so on; but the success of these predictions is evidentially nugatory. In contrast, the idea that water = H₂O naturally suggests that water should be decomposable into hydrogen and oxygen in a 2:1 ratio; this prediction requires no ad hoc posits.

The predictions made by moral theories all appear to require ad hoc posits. The moral theory that killing millions of people is wrong enables us to predict that Hitler won't kill millions of people, only if we make the unmotivated assumption that Hitler is morally admirable (or at least not evil). But the possibility of deriving predictions from an arbitrary combination of moral statements is of no epistemological interest.

Sturgeon gives another example. The pair of claims, 'Injustice causes political instability' and 'Natural property X constitutes injustice' together entail that X causes instability.⁴³ Provided X is independently detectable, this is a testable prediction. Here, Sturgeon's assumption that injustice causes instability does not seem ad hoc; it seems like a natural assumption about the nature of injustice. But why? Presumably because we assume (a) that people can typically identify injustice, and (b) that they object to it. Given these assumptions, when injustice is widespread in a society, it is natural to expect that the society will be politically unstable. If people had no means of moral knowledge to begin with, then there would be no reason to expect that moral properties would be correlated to their behavior in any interesting way.

The problem now is that our ability to make a prediction based on moral claims in this case depends on our assuming that people already have some way of gaining reliable moral beliefs. And what might that be? If we accepted that people have a reliable faculty of moral intuition, say, or that people can observe moral facts, or that they can deduce moral facts from non-moral facts, then we could accept this. But the synthetic reductionist rejects the first possibility, and we have rejected the latter two already. Moreover, if the reductionist could answer this question, then he wouldn't need the present argument—he would already have an account of moral knowledge.

In sum, if we are given all the non-moral properties of a thing, the additional hypothesis that one of those properties constitutes goodness, say, does not seem to naturally generate any new predictions about what we should observe, without a separate ad hoc posit for each desired prediction. Consequently, starting from a system of purely non-moral beliefs, we would not be justified in adding any moral beliefs. There is one exception: it seems natural and not ad hoc to expect that intelligent beings will by and large pursue the good, feel gratified when it is obtained, and so on. But this prediction depends upon the prior assumption that people have some independent means of detecting moral properties, so it is of no use in accounting for the origin of moral knowledge.

***4.4.4 The unifying power of moral explanations**

Perhaps, despite the fact that they only provide explanations for things we already have non-moral explanations for, moral claims can still be justified because they provide better explanations than those formulated using non-moral claims.

Michael S. Moore has developed this suggestion, arguing that moral explanations can achieve a kind of unification not otherwise available.

An analogy will help. We find that a certain peg will not fit into a certain hole. On one level, we could try to explain this fact by citing the trajectories of all the atoms in the block and the chemical and other forces acting on each of them. But a more perspicuous explanation would be: 'This is a square peg, and that's a round hole'. The latter explanation is superior in its unifying power: given this explanation, we know what to expect when we try putting another square peg into another round hole (with a diameter less than the diagonal of the square). The atomic-level explanation misses this higher-level regularity: the next square peg we encounter may be made of a different substance and will certainly have a different arrangement of atoms, so on the atomic level, the same explanation will not apply to it; we will have to give a distinct explanation in each case. In contrast, the 'square peg, round hole' explanation identifies the essential feature common to all these situations.⁴⁴

Now, suppose the property of goodness is identical with some disjunction of natural properties, that is,

$$\text{\$}\text{\$} \text{\textit{Good}} = (F \text{\textit{ or }} G \text{\textit{ or }} H) \text{\$}\text{\$}$$

where F , G , and H are non-evaluative terms. And suppose there is some effect, E , which occurs whenever any of F , G , or H is present. In that case, although the occurrence of E on a given occasion might be explained by citing the presence of F , the better explanation would be the one that cites the presence of goodness. For the latter explanation would achieve a kind of unification the former explanation does not—it enables one to correctly predict what will happen when another instance of goodness occurs.⁴⁵

The crucial question here is, how would one come up with the idea that $\text{\$}\text{\textit{Good}} = (F \text{\textit{ or }} G \text{\textit{ or }} H) \text{\$}$? Presumably, the mere fact that each of several distinct properties brings about a single effect does not justify identifying goodness with the disjunction of those properties. One might be tempted to appeal to common sense beliefs about which things are good—but that would be to revert to a mistake we have earlier criticized. Since at this point the reductionist has not yet successfully explained how people can access moral reality, he is not entitled to assume that our common sense moral beliefs are reliable. Since no advantage of the proposed moral explanation has been identified other than its unifying power, any equally simple explanation that achieves the same unifying power ought to be equally acceptable. Thus, it would apparently make just as much sense to propose that $\text{\$}\text{\textit{Bad}} = (F \text{\textit{ or }} G \text{\textit{ or }} H) \text{\$}$. Or why not invent some other, non-evaluative concept, say 'torfness', and say that $\text{\$}\text{\textit{Torf}} = (F \text{\textit{ or }} G \text{\textit{ or }} H) \text{\$}$ —why not, in other words, say that $(F \text{\textit{ or }} G \text{\textit{ or }} H) \text{\$}$ is identical with a single property, but that property is not evaluative?

Synthetic reductionists concede that the meaning of an evaluative term is not captured by any non-evaluative expression. What the questions of the last paragraph illustrate is that the distinctive, positively evaluative meaning of 'good' plays no role in the sort of explanations the reductionist would offer us—since all that matters is that F , G , and H be brought under a single concept, a negatively evaluative concept or a non-evaluative concept would serve just as well. This stands in contrast with legitimate scientific explanations. When the hypothesis that $\text{\$}\text{\textit{water}} = \text{\textit{H}}_2\text{\textit{O}} \text{\$}$ explains the result of the electrolysis of water, the specific meaning of ' $\text{\textit{H}}_2\text{\textit{O}}$ ' ('a compound of

two hydrogen atoms and one oxygen atom') plays an essential role. One cannot substitute some entirely different concept, such as 'NaCl', in the theory and still produce an equally good explanation—the hypothesis 'water = NaCl' does not explain the experimental result.

4.5 The argument from radical dissimilarity

One argument against ethical reductionism that has received little attention is the argument that, from our grasp of evaluative concepts, we can simply see the falsity of reductionist theories. On the face of it, for example, wrongness seems to be a completely different kind of property from, say, weighing 5 pounds. In brief:

1. Value properties are radically different from natural properties.⁴⁶
2. If two things are radically different, then one is not reducible to the other.
3. So value properties are not reducible to natural properties.

If this argument sounds thin to you, it is probably because you accept one of the following two responses.

First, a reductionist might say that we cannot simply see the nature of a thing merely because we have a concept of it. We could not see, merely from our concept of water, that water was identical with H_2O . But my argument doesn't assume that we can see everything about the nature of those things we have concepts of; it only assumes that we can see some things about them, and that in the case of physical and psychological properties, we can see that they are not evaluative. To illustrate, suppose a philosopher proposes that the planet Neptune is Beethoven's Ninth Symphony. I think we can see that that is false, simply by virtue of our concept of Neptune and our concept of symphonies. Neptune is an entirely different kind of thing from Beethoven's Ninth Symphony. No further argument is needed. Indeed, if a person couldn't see that Neptune is not a symphony, we would say he either had no idea what Neptune was, or had no idea what a symphony was.

Second, a reductionist might say that the argument, if accepted, would also 'refute' the theory that $\text{Water} = \text{H}_2\text{O}$, for 'water' and ' H_2O ' just seem very different on their face. Similarly, the argument would seem to refute the theory that sound consists of compression waves in the air, because 'sound' and 'compression waves' seem very different.

But this is not so. On the face of it, water doesn't seem to be H_2O ; but it is not the case that water seems to not be H_2O . Our pre-scientific concept of water takes it to be a clear, odorless, tasteless, etc., liquid. We cannot, on the basis of this concept, discern anything about what its micro-structure might be like. We have no experience of just seeing that there are no tiny particles composing it, and so on. Nor do we have the sense that water is a different category of thing from H_2O ; on the contrary, we see that they are the same category of thing, namely, physical substances.

I would likewise deny that we have an experience of seeing that sound doesn't consist of compression waves, or that they are different categories of things. We can see that they are the same category of thing: physical events. I suggest that if you have the intuition that sound obviously can't be a compression wave in the air, then it is because by 'sound' you actually mean the experience of sound, which really is a different kind of thing from the compression waves in the air.

The point about the broad category that a thing belongs to is important, because in my view, our having, at least roughly and for the most part, correct intuitions or beliefs about the nature of x is a precondition on our referring to x . I can talk about Plotinus without knowing much about him; about all I know is that he was a philosopher who lived a long time ago. But I could not talk about Plotinus if I had completely inaccurate beliefs (allegedly) about him; for example, if I thought 'Plotinus' referred to a cake, then I would not be talking about Plotinus (the person) at all.[47] Similarly, if 'good' seems to us to refer to something of a fundamentally different category from natural properties, then when we say 'good' we are not talking about a natural property.

4.6 Explaining moral beliefs

I claim to have shown, in sections 4.2 and 4.5, that evaluative properties cannot be reduced to natural properties. In sections 4.3 and 4.4, I have shown that moral knowledge is not empirical. The structure of this latter argument was as follows:

1. Moral claims are known empirically only if they are known by observation or by either deductive or scientific reasoning from observations. (Premise.)
2. Moral claims are not known by observation. (§4.4.1)
3. Moral claims are not known by deduction from non-moral premises. (§4.3)
4. Moral claims are not known by scientific reasoning from non-moral premises. (§4.4)
5. Therefore, moral claims are not known empirically. (From 1, 2, 3, 4.)

I want to make one final, general point about the prospects for reductionism in ethics. The reason why we believe that such phenomena as heat, water, and sound are reducible is that we have compelling scientific arguments for identifying these phenomena with specific theoretical entities. No one doubts the existence of the relevant theoretical entities, because of the systematic power of the scientific theories they belong to in explaining and predicting a host of observable phenomena. Without those theories, no one would believe reductionism about heat, water, or sound.

In ethics, we have no comparable theories in hand. Reductionists typically only predict that moral claims will prove to be reducible.[48] They theorize in the abstract about how it might be possible to justify moral beliefs, but without presenting a detailed argument for any specific moral belief. In fact, moral philosophers have labored for two millennia without

achieving a consensus on any general moral theory. In such a situation, the reductionist's position raises two questions. The first and more obvious is why we should believe that a reduction of moral terms is possible.

The second is how the reductionist explains people's actual moral beliefs. Even if one of the proposals discussed in section 4.4 were viable, none of them corresponds to how we actually form moral beliefs. If 'good' is a theoretical term like H_2O , shouldn't we expect that the concept would be developed only concurrently with the theory of which it is a part—and that identifiable arguments would have been given before that theory was accepted? Any chemist can identify uncontroversial examples of the sort of observations that molecular theory is needed to explain. Why are so many people unable to identify any of the observations that their moral beliefs are needed to explain? Many professionals in the field deny that there are any such observations, and even ethical reductionists will likely admit that the empirical reasoning behind moral theories is subtle and has yet to be clearly articulated. Yet moral beliefs abound in every society in the world, among the young and the old, the educated and the uneducated. Why? Rather than declaring that nearly all moral beliefs are as yet unjustified, shouldn't we look for some much simpler means of gaining moral knowledge?